Impacts of engineering & emerging technology

ENGINEERING WITH MR KELLY

National 5 · Engineering contexts & challenges · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain social, economic and environmental impacts of an engineered product
- ☐ describe how engineering helps limit climate change, linked to greenhouse gases
- ☐ name an emerging technology that is not yet used commercially, and describe it
- ☐ write "explain" answers with one cause and a related effect

I DO — Worked example

Read how a full-mark answer is built

A company is launching an electric bike.

1 Explain one environmental and one social impact of the electric bike. (4 marks)

Full-mark answer each ✓ is one mark

Environmental: the bike produces no exhaust emissions while it is ridden ✓, so air quality in towns improves ✓. Social: the motor makes hills easier, so less fit people can cycle ✓, which improves their health and independence ✓.

Why marks are not awarded

"It is better for the environment and people will like it."

0 marks: no cause and no specific effect for either.

WE DO — Guided practice

Use the starters and word bank to help you

Engineers are working to limit climate change.

2 Describe one way engineers can help limit climate change. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

renewable · producing · energy · electricity · designing · without · generates · Efficiency

Tip: 1 marks = 1 separate, developed points.

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YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which would score as an emerging technology?
☐ Electric cars ☐ Ammonia-fuelled engines ☐ Artificial intelligence ☐ Smartphones

b) Which links a wind farm to tackling climate change correctly?
☐ It is renewable, so it will last for ever ☐ It reduces the pollution caused by cars ☐ It generates electricity without releasing CO_2 ☐ It is quiet and doesn't spoil the landscape

c) "Cheaper running costs for the user" is a…
☐ Social impact ☐ Structural impact ☐ Environmental impact ☐ Economic impact

d) A full "explain" answer needs…
☐ A cause and a related effect ☐ Two causes of the same problem ☐ A list of facts about the topic ☐ A labelled diagram of the system

4 Key terms: write the letter of the matching description next to each term.

1. **Social impact**	____	**A** An effect on people's lives: health, safety, convenience, jobs.
2. **Economic impact**	____	**B** Link to greenhouse gases such as CO_2, not just "pollution".
3. **Environmental impact**	____	**C** An effect on money: costs, savings, jobs, running costs.
4. **Emerging technology**	____	**D** New and not yet in commercial use, e.g. ammonia-fuelled engines, nuclear diamond batteries.
5. **Climate change answers**	____	**E** An effect on the environment: emissions, habitats, resources, waste.

New technologies are being developed by engineers.

5 Name and describe an emerging technology. (2 marks)

A tidal turbine is installed in the Pentland Firth.

6 Explain one economic impact of the tidal turbine on the local area. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?